

DATABASE PROGRAMMING (SECP3623)
2025/2026 – SEM I

QUESTION 1: Library Management System

Author

<u>author_id</u>	name
1	Haruki Murakami
2	Michelle Obama
3	Yuval Noah Harari

Book

<u>book_id</u>	title	genre	authorID
101	Kafka on the Shore	Fiction	1
101	Becoming	Biography	2
102	Sapeins	History	3
103	Norwegian Wood	Fiction	1

Loan

<u>loan_id</u>	<u>book_id</u>	<u>member_name</u>	<u>loan_date</u>	<u>due_date</u>	<u>return_date</u>
L10	101	Ali Ahmad	2025-09-01	2025-09-15	NULL
L11	101	Mei Ling	2025-09-03	2025-09-17	2025-09-10
L12	102	Ravi Kumar	2025-09-05	2025-09-19	NULL
L13	103	Aida Zulkifli	2025-09-06	2025-09-20	2025-09-18

Write SQL statements for each of the following questions:

- Create Book and Loan relations matching the structures shown.
- Rename authorID → author_id in Book relation.
- List book_id, title for books where genre = 'Fiction'.
- Find books with titles containing 'on'.
- Count how many books for each genre.
- Update a book title typo: 'Sapeins' to 'Sapiens'.
- Delete all loans that have a return_date.
- Insert a new author record.
- Make all NULL genres temporarily 'Unknown'.
- List all loans due between '2025-09-15' and '2025-09-19'.
- List books that have never been loaned.

QUESTION 2: Online Food Delivery

Restaurant (rest_id, rest_name, city) **PK: rest_id**

MenuItem (item_id, rest_id, item_name, price, category) **PK: item_id**

Orders (order_id, rest_id, customer, order_time, status) **PK: order_id**

OrderItem(order_id, item_id, qty) **PK: order_id, item_id**

Write SQL statements for each of the following questions:

- a) Add a UNIQUE constraint to prevent duplicate item names values within the same restaurant.
- b) Add CHECK constraints ensuring MenuItem.price > 0 and OrderItem.qty > 0.
- c) Retrieve all menu items whose item_name contains word 'Laksa'.
- d) Increase the price of all items in category 'Drink' by RM0.50.
- e) Update Orders.status to the enumerated set {'Preparing', 'Delivered', 'Cancelled'}.
- f) Identify the most expensive menu item for each restaurant.
- g) List all menu items that have never been ordered.
- h) Count the number of delivered orders for each restaurant.
- i) Compute the number of menu items offered by each restaurant.
- j) Retrieve item_id, item_name and price for the five most expensive items where category = 'Main', sorted by price descending.
- k) Delete any OrderItem rows with qty = 0.

QUESTION 3: University Course Registration

Student

<u>Sid</u>	Sname	Program
AC101	Azli Ismail	SECPH
AC102	Lee Mei Ling	SECRH
AC103	Vinesh A/L Ganesan	SECJH
AC104	Mardihah Kamal	SECVH

Course

<u>Cid</u>	Cname	Credit
SECP3623	Database Programming	3
SECR2414	Operating System	4
SECD2523	Database	3
SECI1143	Statistical Data Analysis	3
SECP2753	Data Mining	3

Enroll

<u>Sid</u>	<u>Cid</u>	<u>Semester</u>	Grade
AC101	SECP3623	202520261	A
AC101	SECI1143	202520261	B+
AC102	SECR2414	202520261	A-
AC102	SECD2523	202520261	A
AC103	SECI1143	202520261	A

Write SQL statements for each of the following questions:

- Create the Enroll table with a composite Primary Key (Sid, Cid, Semester) and Foreign Key constraints.
- Add a GPA DECIMAL(3,2) attribute to the Student table.
- Add a UNIQUE constraint on (Program, Sname) to prevent duplicate student names within the same program.
- Update Vinesh's program to 'SECPH'.
- Insert a new student and enroll the student in two courses within the same semester.
- Retrieve the course(s) that no student has enrolled in.
- List enroll records in semester '202520261' with grade A (any course).
- List students who have taken both 'Database Programming' and 'Database' courses.
- List (Sid, Sname) for students not enrolled in any course.
- Compute each student's total registered credits in semester 202520261.
- Delete enroll records for a student who withdraws (e.g. Sid = 'AC104').